Swastik Sahu

Master of Computer Applications | Batch 2026–28 | **CHRIST (Deemed to be University), Bengaluru**
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PROFESSIONAL SUMMARY

MCA student with experience in full-stack development, REST APIs and AI-enabled applications using Java, Python, JavaScript, React, Node.js and SQL. Skilled in building web applications, integrating backend services and working with GenAI and RAG technologies.

WORK EXPERIENCE & INTERNSHIPS

Research Intern, Amity University, Noida *Dec 2024 – Mar 2025*
- Conducted data analysis and simulations using Python and MATLAB to study physics-based systems and models.
- Processed and interpreted complex datasets, identifying patterns and trends to support research findings.
- Applied statistical and analytical techniques to improve accuracy and reliability of results.

Research Intern, Amity University, Noida *May 2024 – Jul 2024*
- Assisted in research on particle accelerators and their applications, involving analytical modeling and data interpretation.
- Worked with scientific datasets to perform calculations and derive meaningful insights.
- Documented observations and presented findings in a structured format.

PROJECTS

DropoutGuard AI — Early-Warning & Intervention System | React, Node.js, Python, ML *Present*
- Developed an AI-powered system analyzing academic and engagement indicators to identify students at risk of dropout and support timely intervention.
- Built an explainable Random Forest model using the UCI Student Dropout dataset (4,424 records), achieving 86.78% accuracy and 0.9251 ROC-AUC.
- Implemented risk factors, What-If intervention simulation and faculty workflows using React, Node.js/Express, Python/FastAPI, MongoDB and Scikit-learn.

LetsCollab — Collaborative Project Management Platform | Next.js, Node.js, Socket.IO *2026*
- Built a real-time project management platform with live task, comment, and notification sync using Socket.IO and presence indicators.
- Implemented optimistic concurrency control with HTTP 409 conflict detection and a diff dialog for resolving stale edits.
- Enforced role-based access control (Admin, Project Manager, Team Member, Viewer) at the API layer, with immutable audit logging.
- Developed the stack with Next.js 14, React, TypeScript, Node.js, Express, Prisma, and JWT authentication, with Jest and Supertest.

SKILLS

Languages: Python, Java, JavaScript, SQL **Frontend:** HTML, CSS, Tailwind CSS, React.js, Next.js, Vite
Backend: Node.js, Express.js, REST APIs **Databases:** MongoDB, Mongoose, Oracle SQL
Tools: Git, GitHub, Vercel **Spoken Languages:** English – Proficient (C1) | Hindi – Fluent | Odia – Fluent

CERTIFICATIONS & TRAINING

- **Winter School on Generative AI** – IIT Jodhpur *Dec 2024*
- **Foundations of AI Fundamentals** – IBM *May 2026*
- **Applications of Machine Learning in Urban Studies** – ISRO *Jun 2023*
- **Google AI Essentials Specialization** – Google *May 2026*

CO-CURRICULAR & LEADERSHIP

Class Representative, CHRIST University, Bengaluru *Jun 2026 – Present*
Infobahn, CHRIST University, Bengaluru *2026 – Present*
Editorial Team Member
International Young Physicists' League, IAPS *Feb 2024 – Mar 2024*
Organizing Committee
Karuna The Compassion NGO, New Delhi *Dec 2023 – Mar 2024*
Volunteer

ACHIEVEMENTS & AWARDS

- **Hack Devengers 2.0 – Top 50 Projects** – Team Devengers *Sep 2026*
- **Poster Presentation** – National Symposium on Advanced Research of Plasma Science Society, PDEU, Gujarat *Dec 2024*
- **Science Hackathon 2023** – Indian International Science Festival by Hack2Skill *Sep 2023*

EDUCATION

CHRIST (Deemed to be University), Bengaluru *2026 – 2028*
Master of Computer Applications (MCA) | GPA: 3.55 / 4.0

Amity University Uttar Pradesh, Noida *May 2025*
B.Sc. Physics (Hons.) | GPA: 8.89 / 10.0